

Variations of Gauss' Circle Problem

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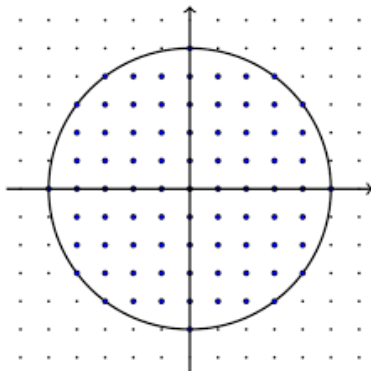
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Motivation

We begin with looking at the classic Gauss' Circle problem, which counts the number of points on an integer lattice inside a circle of some radius. Define:

$$r_2(n) = \#\{a, b \in \mathbb{Z}^+ \mid a^2 + b^2 = n\}$$



A Preliminary Estimate

Theorem (Gauss)

As $x \rightarrow \infty$ we have

$$\sum_{n \leq x} r_2(n) = \frac{\pi x}{4} + O(\sqrt{x})$$

We also introduce the following, as to motivate future definitions:

Definition

$$r_2(n) = \sum_{d|n} \chi_4(d)$$

where $\chi_4(d)$ is the non-principal Dirichlet character modulo 4, i.e.

$$\chi_4(p) = \begin{cases} 0, & p = 2 \\ 1, & p = 1(4) \\ -1 & p = 3(4) \end{cases}$$

Definition

We now define another arithmetic function.

$$\omega_*(n) = \sum_{\substack{p|n \\ p \equiv 1(4)}} 1$$

The goal of introducing this is to utilize a dispersion method to study $r_2(n)$ in more depth and obtain a better error.

Theorem

As $x \rightarrow \infty$ we have:

$$\sum_{n \leq x} \omega_*(n) = \frac{1}{2}x \log \log x + Ax + O\left(\frac{x}{\log x}\right)$$

where $A = 0.26\dots$ is Meissel-Mertens constant.

and in general we have (as $x \rightarrow \infty$)

$$\sum_{n \leq x} \omega_*^k(n) \sim \frac{1}{2^k} x (\log \log x)^k$$

which follows from the values of omega being normally distributed, as is discussed in more depth later.

Theorem

As $x \rightarrow \infty$ we have:

$$\sum_{n \leq x} r_2(n) \omega_*(n) = \frac{\pi x}{4} \log \log x + O(x)$$

and

$$\sum_{n \leq x} r_2(n) \omega_*^2(n) = \frac{\pi x}{4} (\log \log x)^2 + O(x \log \log x)$$

One way to interpret this is that we can essentially multiply our individual asymptotic formulae to obtain a main term for their multiplication before summation.

Proof

Let $S(x)$ denote the sum in question.

$$\begin{aligned} S(x) &= \sum_{n \leq x} r_2(n) \omega_*(n) = \sum_{n \leq x} r_2(n) \sum_{\substack{p|n \\ p \equiv 1(4)}} 1 \\ &= \sum_{\substack{p \leq x \\ p \equiv 1(4)}} \sum_{m \leq x/p} r_2(mp) \\ &= \sum_{\substack{p \leq x \\ p \equiv 1(4)}} \sum_{m \leq x/p} 2r_2(m) - r_2(m/p) \\ &= \sum_{\substack{p \leq x \\ p \equiv 1(4)}} 2 \left(\frac{\pi}{4} \frac{x}{p} + O(\sqrt{x/p}) \right) - \sum_{\substack{p \leq x \\ p \equiv 1(4)}} \left(\frac{\pi}{4} \frac{x}{p^2} + O\left(\frac{\sqrt{x}}{p}\right) \right) \end{aligned}$$

From here it is a simple evaluation using established analytic nt tools.

Proof (2)

Simply note that

$$\sum_{\substack{p \leq x \\ p \equiv 1(4)}} \frac{1}{p^2} \ll \sum_{n=1}^{\infty} \frac{1}{n^2} \ll 1, \quad \sum_{\substack{p \leq x \\ p \equiv 1(4)}} \frac{1}{p} \ll \log \log x$$

and we obtain

$$\begin{aligned} S(x) &= \frac{\pi x}{2} \sum_{\substack{p \leq x \\ p \equiv 1(4)}} \frac{1}{p} + O\left(\sqrt{x} \sum_{\substack{p \leq x \\ p \equiv 1(4)}} \frac{1}{\sqrt{p}}\right) + O(x) \\ &= \frac{\pi x}{2} \sum_{\substack{p \leq x \\ p \equiv 1(4)}} \frac{1}{p} + O(x) = \frac{\pi x}{4} \log \log x + O(x) \end{aligned}$$

Application of Dispersion Method

We want to now compute

$$\mathcal{D}(x) = \sum_{n \leq x} r_2(n)(\omega_*(n) - \log \log x)^2$$

since this can tell us more about the behavior of $r_2(n)$.

$$\begin{aligned}\mathcal{D}(x) &= \sum_{n \leq x} r_2(n)(\omega_*(n) - \log \log x)^2 \\ &= \frac{\pi x}{4}(\log \log x)^2 - 2 \log \log x \cdot \frac{\pi x}{4} \log \log x + (\log \log x)^2 \frac{\pi x}{4} + O(x \log \log x) \\ &= O(x \log \log x)\end{aligned}$$

In particular, this says that large values of $r_2(n)$ come from numbers with $\omega_*(n) \sim \log \log n \sim \log \log x$

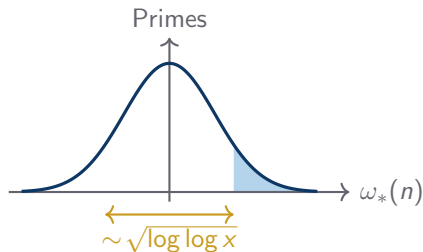
Intuition: Why Large r_2 Values Force $\omega_* \sim \log \log x$

Key insight from the computation:

- Large values of $r_2(n)$ come from n 's with

$$\omega_*(n) \sim \log \log x.$$

- Such an n has primes $\equiv 1 \pmod{4}$ appearing twice as often as in a random integer.



$$\frac{\omega_*(n) - \frac{1}{2} \log \log n}{\sqrt{\log \log x}} \geq a$$

Erdős-Kac Theorem tell us this is normally distributed

Where to go?

There are a few directions we can go from here.

1. Change r_2 to something else, like $\sigma(n) = \sum_{d|n} d$
2. Improve on $\mathcal{D}(x) \ll x \log \log x$ to understand the error term
3. Restrict $r_2(n)$ to something like $r_1(n) := \#\{a \in \mathbb{Z}^+, p \in \mathbb{P} \mid a^2 + p^2 = n\}$
4. Take more squares, i.e. $r_4(n) := \#\{a, b, c, d \in \mathbb{Z}^+ \mid a^2 + b^2 + c^2 + d^2 = n\}$

More Squares

The direction we initially explored was taking more squares. We find:

Theorem

As $x \rightarrow \infty$

$$\sum_{n \leq x} r_4(n) \omega_*(n) = \frac{\pi^2}{12} \frac{15}{16} x^2 \log \log x + O(x),$$

$$\sum_{n \leq x} r_4(n) \omega_*^2(n) = \frac{\pi^2}{12} \frac{15}{16} x^2 (\log \log x)^2 + O(x^2 \log \log x)$$

as an initial estimate, but one can actually do even better in the ω_*^2 case:

Lemma

$$\sum_{n \leq x} r_4(n) \omega_*(n) = Cx^2 (\log \log x)^2 + CAx^2 \log \log x + O(x^2)$$

More Squares (2)

It is intuitive to think that $r_4(n)$ should be related to $r_2(n)$ in some way, and indeed we have:

$$r_4(n) = \sum_{m \leq n} r_2(m)r_2(n-m)$$

Additionally, following the intuition of a circle for r_2 , one can think of $\sum r_4$ as counting points inside $\mathcal{S}^3$.

Open Problem

We can again vary the number of squares.

$$n = a^2 + b^2 + c^2 + d^2 \text{ Known - Lagrange}$$

$$n = p^2 + b^2 + c^2 + d^2 \text{ Known}$$

$$n = p^2 + q^2 + c^2 + d^2 \text{ Known - Dispersion Method}$$

Other - Open

Restriction

As proposed by [Daniel, 2001]:

Theorem (Daniel)

As $x \rightarrow \infty$ one has:

$$\sum_{n \leq x} r_1^2(n) = \left(\frac{\pi}{2} + \frac{9}{4} \right) \frac{x}{\log x} + O\left(\frac{x(\log \log x)^2}{(\log x)^2} \right)$$

however in his proof, he uses a suboptimal version of a well known theorem (BDH) that actually has a better bound. Thus, we conjecture

Conjecture

$$\sum_{n \leq x} r_1^2(n) = \left(\frac{\pi}{2} + \frac{9}{4} \right) \frac{x}{\log x} + C \frac{x}{\log^2 x} + o\left(\frac{x}{(\log x)^2} \right)$$

Summary

- We can study asymptotic formulae via dispersion method
- Erdős-Kac ensures that the distribution is normal to study r_2
- Integers weighted by r_2 contain on average twice the expected $p \equiv 1(4)$ factors
- Outlook: Find better asymptotic formulae for variations of $r(n)$

The End